

EXECUTIVE SUMMARY

Business Problem

Derm AI Diagnostics required a structured analysis of patient and skin lesion data to understand the demographic, clinical and anatomical characteristics represented in the dataset and identify characteristics that differed across lesion diagnoses and broader benign, pre-cancerous and malignant categories. Particular attention was given to the characteristics that distinguished malignant from benign lesions and to data quality issues that could affect interpretation.

Method

The analysis was conducted in PostgreSQL using two source tables containing patient-level and lesion-level information for 1,088 patients and their associated lesions. Data quality was assessed before analysis, with validated categorical inconsistencies standardised while unresolved zero values in Fitzpatrick skin type and lesion diameter variables remained untouched.

Cleaned patient and lesion data were linked through *patient_id* to create a unified analytical view. Descriptive analysis was first conducted across individual diagnoses and subsequently across benign, pre-cancerous and malignant lesion categories. Frequencies, proportions and grouped comparisons were used to examine demographic characteristics, clinical features, anatomical distribution, clinical history, lifestyle and environmental characteristics.

Key Findings

Pre-cancerous lesions were the most common category, accounting for 42.37% of lesions, followed by malignant lesions at 31.80% and benign lesions at 25.83%. Malignant lesions showed the broadest clinical feature profile, with bleeding (52.02%) and pain (36.99%) particularly pronounced compared with benign lesions (1.42% and 1.78%, respectively).

Anatomical patterns also differed: the face was the most common site for malignant and benign lesions, while pre-cancerous lesions were concentrated on the forearm. Patients with malignant and pre-cancerous lesions were predominantly older, whereas patients with benign lesions were represented across a broader age range.

Clinical history provided another notable distinction. Among patients with malignant lesions, 54.05% had a family history of cancer and 45.38% had a previous skin cancer diagnosis, compared with 6.41% and 4.63%, respectively, among patients with benign lesions.

Overall, no single characteristic distinguished the lesion categories. Rather, the clearest differences emerged from combinations of lesion-level clinical characteristics and patient-level demographic and clinical history characteristics.

Limitations

The findings are descriptive and should not be interpreted as causal relationships or validated clinical predictors. Unequal diagnostic group sizes, particularly the small number of melanoma cases, require caution when comparing percentages. Interpretation of Fitzpatrick skin type and lesion diameter was restricted by unresolved zero values, while substantial unknown values for *background_father* and *background_mother* limited meaningful analysis of parental background associated with lesions.

Recommendations

Future analysis should prioritise combinations of clinical, demographic and patient history characteristics rather than relying on individual characteristics in isolation. Data documentation should also be strengthened, particularly for Fitzpatrick skin type and lesion diameter coding, before these variables are incorporated into further analysis. The descriptive patterns identified here provide a basis for more focused statistical or predictive investigation where clinically appropriate.

Next Steps

The next analytical stage should clarify the unresolved variable definitions with the data provider and, where appropriate, investigate whether the combinations of characteristics identified in this analysis can reliably distinguish between lesion categories using more advanced statistical or predictive methods.